Fact-checking RL project: team brief

Week of 8 Oct 2026 · Project ends 11 Nov 2026 · Agentic lead: Chia Kwang Yang

Background

A fact-checker that answers confidently when the evidence is unclear spreads false confidence. Our system takes a claim and returns three things: a verdict (Supported, Refuted or Not Enough Info), the Wikipedia sentences it relied on, and a confidence. On 900 FEVER claims, Qwen3.5-2B is right only 47% of the time even with the correct evidence in front of it; it calls almost everything Supported. Fine-tuning (SFT) lifts it to 78%, above a prompted 180B model (72% with our retrieval); it writes confidence 1.0 every time, but its verdict probability is well calibrated (error 0.03). Our first GRPO run (no KL penalty) gained nothing: the model grew over-certain, raising that error to 0.21. **Decided:** we train the 2B; the 4B was only about 1 point better after SFT.

What we aim to show by 11 November

1. **RL improves the verifier without breaking calibration.** The GRPO-trained 2B verifier beats the SFT model on accuracy or harm-weighted error (calling a false claim true is the worst mistake) while its verdict probability stays calibrated (error at most 0.05), on the same 900 claims, and holds up on the shortcut tests.
2. **A cost-aware controller.** A small learned policy that decides when to search again, verify, abstain or answer sits above fixed policies on the accuracy-versus-search-cost curve.
3. **Multi-turn vs single-turn RL,** same model and reward; only the loop differs.
4. **Transfer and demo.** Results on SciFact and AVeriTeC, and a live demo that shows the decision trace.

Who does what

Person, role	This week (8 Oct)	Weeks 2–4 (15 Oct – 1 Nov)	Done when
RL group: first GRPO run done (no gain, calibration lost). Confidence is now the verdict probability, not the written number.			
Chao Chien Hung RL-1 verifier	Rerun GRPO with the KL penalty on ($\beta \approx 0.04$), on the 926 claims whose 8 samples score differently; stop at 200 steps if calibration slips (GPU)	Wk 2: final verifier, 3 seeds; fallback: SFT that writes its verdict probability (GPU). Wk 3: before/after tables. Wk 4: FEVER+SciFact vs FEVER-only run, scored on SciFact dev (GPU)	GRPO beats SFT on accuracy or harm, calibration error $\leq$0.05; SFT vs GRPO table
Michel Yildirim RL-2 reward	Temperature-scaling baseline on the verdict probability (the bar RL must beat); calibration metrics in AG-3's eval	Wk 2: 4 ablations on the fixed setup, overnight (GPU); hacking checks (confidence tracks correctness per claim, not per label). Wk 3: reliability diagrams. Wk 4: buffer	Reliability diagrams before/after; ablation table; hacking checks pass
Yao Tianhao RL-3 controller	Controller env on real FEVER; abstain confidence = verifier's NEI probability	Wk 2: controller training (CPU). Wk 3: cost sweep; fixed-policy baselines. Wk 4: buffer	Accuracy-vs-cost curve; controller vs fixed policies
Agentic group: the loop, the tools, the data and the demo			
Yu Jing AG-1 retrieval	Build the `search(query, k)` tool	Wk 2: sentence recall; Exa search for AVeriTeC (date cutoff, fact-check sites excluded, cached), 50-claim cost pilot. Wk 3: `follow()` hyperlink tool on FEVER; re-query	Retrieval recall table across datasets
Chia Kwang Yang AG-2 agent loop	Loop skeleton over all 900 claims; one multi-turn GRPO test (GPU); keep the GPU calendar	Wk 2: prompted agentic baseline (GPU); loop streams each step as an event (FastAPI). Wk 3–4: multi-turn GRPO runs 1 and 2 (GPU); GRPO verifier into the demo	Multi-turn vs single-turn GRPO on the 900 claims; live event stream
Bryan Wu Jiahe AG-3 eval & demo	Eval script and frozen test splits	Wk 2: shortcut test set; loaders; demo Browse mode on saved traces. Wk 3: failure audit in Browse; Live mode. Wk 4: transfer runs; Compare mode	Every reported number comes from the eval script; demo with fallback

How we work together

- **Freeze four interfaces this week** so nobody waits on anyone: `search()` (AG-1), `verify()` (RL-1 trains it, AG-2 sets the signature), `reward()` (RL-2), and the trace JSONL, one row per claim (AG-2 writes it; RL-3 trains on it; AG-3 scores it).
- **No number goes on a slide unless it came from AG-3's eval script.**
- **One GPU** (RTX 5070 Ti, 16GB). Week 1: RL-1 by day; one AG-2 test. Week 2: RL-1 final verifier by day; AG-2 baseline and RL-2 ablations share nights. Week 3: AG-2 first, leftover ablations when free. Week 4: AG-2 by day, RL-1 SciFact run overnight. Week 5: spare. Book with KY.
- **Reviews:** 15 Oct (does fixed GRPO beat SFT and keep calibration?), 29 Oct (pick the lead claim), 5–11 Nov (report, slides, demo dry run, final review).
- **Demo:** one frontend fed by the loop's trace events. *Live*: any claim, steps appear as they run (Exa for open claims). *Compare*: SFT vs GRPO on the same claim. *Browse*: saved traces; doubles as the failure audit and the fallback.
- **Tools by dataset:** FEVER uses Wikipedia search plus hyperlink follow; SciFact searches its abstracts; AVeriTeC and the demo use Exa.
- **Open:** AVeriTeC's fourth label (Conflicting/Cherrypicking) mapped to NEI or dropped, AG-3 to decide. Exa key: KY. Out of scope: dense retriever, escalate action, LIAR-PLUS.